

Phishing Alert Response

By Derell Facey

*THIS PROJECT WAS AN ACTIVITY FROM THE “*Google Cybersecurity Professional Certificate*”*

Project Overview:

Scenario: You are a level-one SOC analyst at a financial services company and you’ve received a phishing alert about a suspicious file being downloaded on an employee’s computer. Using your company’s Phishing Playbook, follow the process for investigating and resolving the alert.

Tools:

- [VirusTotal](#)
 - [Phishing Playbook](#)
-

1) Examine the Alert Ticket:

Alert ticket information:

Ticket ID	Alert Message	Severity	Details	Ticket status
A-2703	SERVER-MAIL Phishing attempt possible download of malware	Medium	The user may have opened a malicious email and opened attachments or clicked links.	Open ▾

Additional information

Known malicious file hash:

54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b

Email:

From: Def Communications <76tguyhh6tgftrt7tg.su> <114.114.114.114>

Sent: Wednesday, July 20, 2022 09:30:14 AM

To: <hr@inergy.com> <176.157.125.93>

Subject: Re: Infrastructure Egnieer role

Dear HR at Inergy,

I am writing for to express my interest in the engineer role posted from the website.

There is attached my resume and cover letter. For privacy, the file is password protected. Use the password paradise10789 to open.

Thank you,

Clyde West

Attachment: filename="bfsvc.exe"

The email contains the following signs that may indicate a phishing attack:

- Severity = Medium indicates the ticket may need to be escalated.
- Mismatched sender email and sender name:
 - From: Def Communications
 - Email: 76tguyhh6tgftrt7tg.su
- Grammatical errors:
 - “Subject: Re: Infrastructure Egnieer role”
 - “I am writing for to express...”
 - “There is attached my resume and...”

2) Investigate attachments:

Upon examination, we change the ticket status to “investigating”, and analyze the suspicious attachment.

Ticket ID	Alert Message	Severity	Details	Ticket status
A-2703	SERVER-MAIL Phishing attempt possible download of malware	Medium	The user may have opened a malicious email and opened attachments or clicked links.	Investigating ▾

Known malicious file hash:

54e6ea47eb04634d3e87fd7787e2136ccfbcc80ade34f246a12cf93bab527f6b

Attachment: filename="bfsvc.exe"

We can determine if the file is malicious by searching the file hash in a file scanning tool such as VirusTotal:

Upon analysis, the file seems to be malicious with **60/72** vendors flagging it.

60

/ 72

Community Score

-247

60/72 security vendors flagged this file as malicious

Reanalyze Similar More

54e6ea47eb04634d3e87fd787e2136ccfbcc80ade34f246a12cf93bab527f6b

bfsvc.exe

Size430.00 KB

Last Analysis Datea moment ago



peexe

direct-cpu-clock-access

long-sleeps

service-scan

runtime-modules

spreader

checks-user-input

detect-debug-environment

Activity Summary

Download Artifacts Full Reports Help

⚠ Detections

3 MALWARE

🏳️ Mitre Signatures

2 LOW 38 INFO

🛡️ IDS Rules

1 LOW

🔍 Sigma Rules

NOT FOUND

📁 Dropped Files

757 OTHER 1 PE_EXE 1 TEXT

1 JPEG 1 HTML 1 DOC

1 XML

🌐 Network comms

36 HTTP 104 DNS 823 IP

3 JA3

Behavior Tags

checks-user-input crypto detect-debug-environment direct-cpu-clock-access long-sleeps persistence runtime-modules self-delete service-scan

Dynamic Analysis Sandbox Detections

⚠ The sandbox Yomi Hunter flags this file as: MALWARE

⚠ The sandbox DAS-Security Orcas flags this file as: MALWARE

⚠ The sandbox CAPE Sandbox flags this file as: MALWARE

Individual security vendor analyses:

[illegible]

Additionally, the file “bfsvc.exe” is not signed, further indicating that the file is malicious.

Signature info ⓘ

Signature Verification

⚠ File is not signed

File Version Information

Copyright	© Microsoft Corporation. All rights reserved.
Product	Microsoft® Windows® Operating System
Description	Boot File Servicing Utility
Original Name	bfsvc.exe
Internal Name	bfsvc.exe
File Version	6.1.7601.17514 (win7sp1_rtm.101119-1850)

The file contacted several domains, IP addresses, and URLs that are flagged as malicious. One such malicious domain was `misecure.com` :

6 / 94

Community Score -57

⚠ 6/94 security vendors flagged this domain as malicious

misecure.com

phishing and fraud Phishing and Other Frauds top-100K dynamic-dns

Registrar PDR Ltd. d/b/a PublicDomainRegistry.com

Creation Date 23 years ago

Last Analysis Date 19 hours ago

Reanalyze Similar More

Since we’ve confirmed that the email contains a malicious attachment, we set the ticket status to “Escalated” and notify a level-two SOC analyst of the ticket.

Ticket ID	Alert Message	Severity	Details	Ticket status
A-2703	SERVER-MAIL Phishing attempt possible download of malware	Medium	The user may have opened a malicious email and opened attachments or clicked links.	Escalated ▾

3) Summary of Incident:

- Who caused the incident?
 - A malicious actor posing as a candidate for a Infrastructure Engineer role
- What happened?
 - A phishing email was used to successfully compromise a company computer.
 - A malicious file attachment was downloaded and executed.
- When did the incident take place?
 - Wednesday, July 20, 2022 at 09:30:14 AM.
- Where did the incident occur?
 - At a financial services company.
- Why did it happen?
 - An employee at HR downloaded and executed the file attachment in the phishing email.